

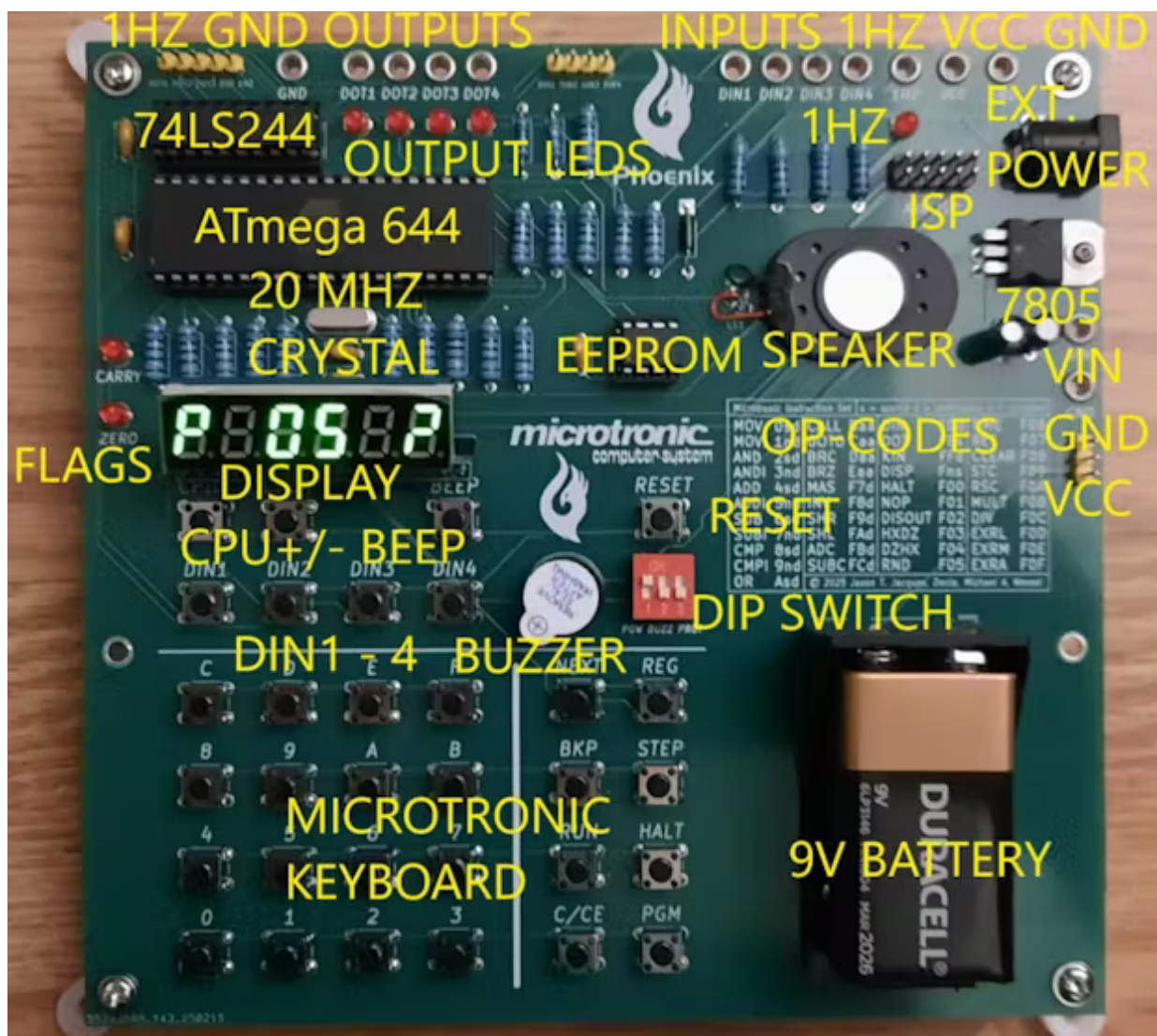


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Makers and vintage computing enthusiasts Michael Wessel, Jason T. Jacques, and Decle have built the first Busch Microtronic since the 1980s, running a copy of the recently-recovered original ROM in emulation: the Microtronic Phoenix.

"In a previous effort we have successfully recovered the original firmware ROM of the [Busch] Microtronic," Wessel explains of the team's work. "With the original firmware available, it is now time to let it fly again on modern hardware, resulting in the first fully authentic re-implementation of the Busch Microtronic from 1981! Rise from your ashes, Microtronic!"



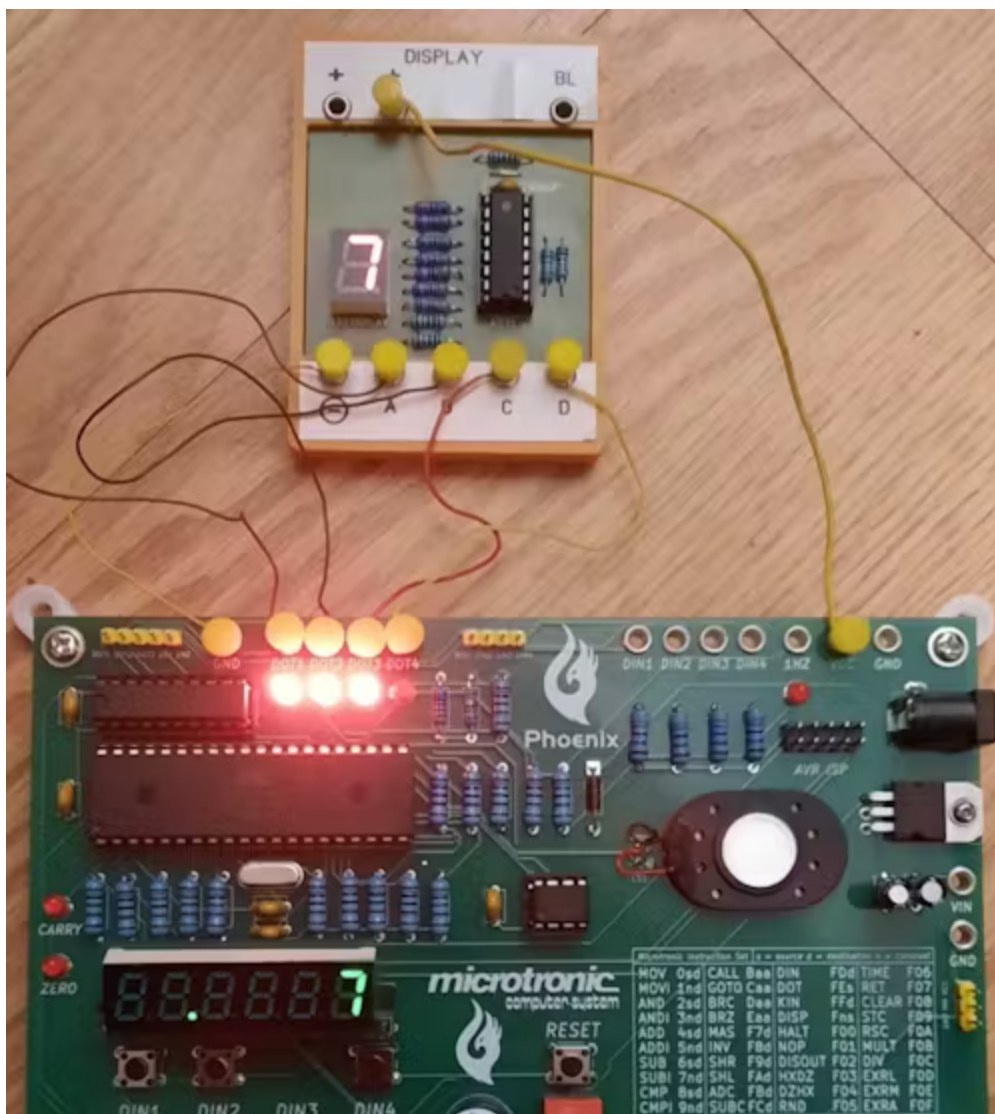
Got a hankering for a little four-bit retro computing? Build yourself a Microtronic Phoenix. (📷: Michael Wessel)

Released in 1981, the Busch 2090 Microtronic Computer System — to give the device its proper title — was an unusual microcomputer trainer built by Busch Modellbau, a company founded as a fireworks manufacturer before pivoting to plastic models. The original used a Texas Instruments TMS1600 four-bit processor running at just 500kHz and with a mere 576 bytes of RAM and 4kB of ROM — and it's the contents of that ROM, long thought lost, that Wessel and friends were able to recover.

While there are hard- and software emulators of the machine available, none of them were running the original ROM — until the launch of the Microtronic Phoenix, a hardware recreation running an emulator on a Microchip ATmega644P-20U microcontroller — and eight-bit part running at a considerably faster 20MHz. Elsewhere on the board is a 24LC256 EEPROM as

a non-volatile storage device and a 74LS244 to protect the microcontroller's pins from the machine's input/output ports and vice-versa. "In principle," Wessel explains, "the Phoenix is a one-chip design — everything is driven by the ATmega644P-20U, even the display. The 24LC256 EEPROM is optional; the machine will also work without it."

The emulator, which includes the choice of using six-digit seven-segment LED or bubble LED display or modern equivalent, includes a physical keyboard, buzzer, and silkscreen instruction set reference, and no fewer than three firmware options: Neo Mode, a reimplement of the firmware; Phoenix Mode, which runs the original Busch ROM; and Combined, which allows either to be selected on startup.



The recreation even includes functional general-purpose input/output pins, compatible with original Microtronic accessories. (📷: Michael Wessel)

"In Neo Mode," Wessel explains, "the hardware emulator has access to the additional hardware features on the board: a speaker for sound output, a 256kBit 24LC256 EEPROM for mass-storage of Microtronic RAM dumps, instead of the 2095 cassette interface; [and a] seven-segment status display."

The project is documented in full on Hackaday.io (<https://hackaday.io/project/202835-microtronic-phoenix/>); board Gerber files and firmware source code are available on GitHub (<https://github.com/lambdamikel/microtronic-phoenix?tab=readme-ov-file>) under the reciprocal GNU General Public License 3.

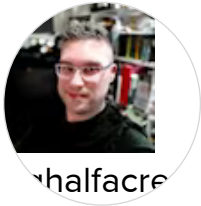
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Gareth Halfacree (/ghalfacree) ✓

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